



THE 2024 ICPC ASIA PACIFIC CHAMPIONSHIP

VNU UNIVERSITY OF ENGINEERING AND TECHNOLOGY

2ND MARCH 2024



Problem C

Bit Counting Sequence

Time limit: 2 seconds

For a non-negative integer x , let $p(x)$ be the number of ones in the binary representation of x . For example, $p(26) = 3$ because $26 = (11010)_2$.

You are given a sequence of n integers $(a_1, a_2, \dots, a_n)$. Your task is to determine whether there exists a non-negative integer x such that $(p(x), p(x+1), \dots, p(x+n-1))$ is equal to $(a_1, a_2, \dots, a_n)$. Furthermore, if it exists, compute the smallest x satisfying the condition.

Input

The first line of input contains one integer t ($1 \leq t \leq 1000$) representing the number of test cases. After that, t test cases follow. Each of them is presented as follows.

The first line contains one integer n ($1 \leq n \leq 500\,000$). The second line contains n integers $a_1, a_2, \dots, a_n$ ($0 \leq a_i \leq 60$ for all i).

The sum of n across all test cases in one input file does not exceed 500 000.

Output

For each test case, output the smallest non-negative integer x satisfying the condition above. If there is no such x , output -1 instead.

Sample Input #1

Sample Output #1

4	13
5	3
3 3 4 1 2	2305843009213693949
3	-1
2 1 2	
2	
60 60	
2	
8 0	

Explanation for the sample input/output #1

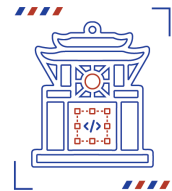
For the first test case, $x = 13$ satisfies the condition above since $(p(13), p(14), p(15), p(16), p(17)) = (3, 3, 4, 1, 2)$. It can be shown that there is no non-negative integer smaller than 13 that satisfies the condition above.



THE 2024 ICPC ASIA PACIFIC CHAMPIONSHIP

VNU UNIVERSITY OF ENGINEERING AND TECHNOLOGY

2ND MARCH 2024



This page is intentionally left blank.